

Romain Verdière

PhD in Applied Mathematics

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Research experience

- 2025–2026 **Temporary Teaching and Research Assistant (ATER)**, *Université Grenoble Alpes*, Grenoble
- Diffeomorphism based feature learning at scale
 - Density estimation using sparse transport maps
 - Surrogate modeling for the Eco3M-Med Model
- 2022–2025 **PhD Candidate in Applied Mathematics**, *INRIA*, Grenoble
- Linear and nonlinear dimension reduction methods for function approximation.
Supervisors: Clémentine Prieur, Olivier Zahm.
- March–August 2018 **Master’s Thesis Internship**, *LaBRI*, Bordeaux
- Routing algorithms in ad-hoc networks of connected objects (IoT).
Supervisor: Damien Magoni.
- May–July 2016 **Research Internship**, *Trinity College*, Dublin
- Sequential Monte Carlo methods for real-time estimation of driving parameters in intelligent vehicles.
Supervisors: Julien Monteil, Mélanie Bouroche.

Scientific contributions

Journal papers

- 2025 **Diffeomorphism-based feature learning using Poincaré inequalities on augmented input space**, *Romain Verdière, Clémentine Prieur, Olivier Zahm*, *Journal of Machine Learning Research (JMLR)*
- 2025 **Mollified Active Subspace**, *Romain Verdière, Clémentine Prieur, Olivier Zahm*, Preprint

Oral presentations

- 2026 **LBMC, ENS de Lyon, Lyon**, Gradient-based dimension reduction
- 2026 **Aptikal’s seminar, LIG, Grenoble**, Gradient-based dimension reduction
- 2025 **SAMO, RT-UQ PhD Day, Grenoble**, Mollifiers to enhance gradient-based dimension reduction
- 2024 **CIROQUO Training – Feature Extraction, Grenoble**, Diffeomorphism-based feature learning
- 2024 **Journées de la Statistique, Bordeaux**, Diffeomorphism-based feature learning
- 2024 **SIAM UQ, Trieste, Italy**, Diffeomorphism-based feature learning

Poster sessions

- 2024 **MASCOT-NUM, Sophia Antipolis**, Diffeomorphism-based feature learning, Best Poster Award
- 2023 **CIROQUO Consortium, Palaiseau, France**, Diffeomorphism-based feature learning

Teaching experience

- 2025–2026 **Temporary Teaching and Research Assistant (ATER)**, *Université Grenoble Alpes*, Grenoble, 184h
Lectures, tutorials, and practical sessions in calculus, elementary algebra, and probability/statistics for first- and second-year undergraduate students.
- 2022–2024 **Adjunct lecturer**, *La Prépa des INPs*, Grenoble, 90h
Tutorials and lectures for second-year probability and statistics course: discrete and continuous probability, convergence of random variables, statistical estimators, confidence intervals.
- 2021–2022 **Oral examiner (CPGE MPSI and MP*)**, *Lycée Camille Guérin*, Poitiers, 4h/week
- 2021–2022 **Mathematics teacher**, *Collège Théophraste Renaudot*, Saint-Benoît (Vienne), Full time, 20h/week
Grades: 6th, 7th, and 8th grade.
- 2020–2021 **Oral examiner (CPGE PCSI and MP)**, *Lycée Michel de Montaigne*, Bordeaux, 2h/week
- 2020–2021 **Trainee mathematics teacher**, *Lycée Alfred Kastler*, Talence, 8h/week
Grades: 10th grade, 11th grade.
- 2009–2012 **Programming workshop instructor**, *Jeunes-Science Bordeaux*, Bordeaux, 2 weeks/year

Industry experience

- February–July 2017 **Gap year internship**, *Thales Avionics*, Toulouse
Simulation of an obstacle-detection radar system for low-altitude helicopter flight.
- September–January 2017 **Gap year internship**, *Amadeus IT Group*, Sophia Antipolis
Development of a tool for analyzing flight quality criteria in commercial tests.
- July 2015 **Technical internship (engine mechanic)**, *AIA Bordeaux*

Education

- 2022–2025 **PhD in Applied Mathematics**, *Université Grenoble Alpes*
- 2019–2020 **Agrégation in Mathematics**, *Université Bordeaux 1*
- 2018–2019 **MSc year 1 – Fundamental Mathematics**, *Université Bordeaux 1*
- 2014–2018 **Engineering degree**, *ENSTA Paris*, Palaiseau, Track: Information Systems Architecture and Security
- 2012–2014 **Preparatory classes (CPGE MPSI/PSI*)**, *Lycée Michel de Montaigne*, Bordeaux
- 2009–2012 **French Baccalaureate**, *Lycée des Graves*, Gradignan, Scientific track

Skills

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| Mathematics | Probability and Statistics, Approximation Theory, Statistical Learning, General and Applied Mathematics |
| Machine Learning | PyTorch, Scikit-learn, NumPy, Python, Matlab |
| Programming | Python, C, C++/Qt, Julia, SQL, Java, PHP, HTML/CSS |
| Linux | General system knowledge, Bash, Networking, Virtual Machines, Git |
| Writing | LaTeX, Beamer |
| Languages | French (native), English (C1) |
| Driving License | Category B |

Extracurricular activities

- Sports
 - Rock climbing (single-pitch, multi-pitch, bouldering)
 - Skiing, ski touring, road cycling, hiking
- DIY
 - Arduino, Raspberry Pi
 - Conversion of a cargo van into a certified camper van (VASP)